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A multi-perspective study on Artificial Intelligence in Education: grants, conferences, journals, software tools, institutions, and researchers

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ABSTRACT

With the rapid development of artificial intelligence (AI) technologies and a continuously growing interest in their application in educational contexts, there has been significant growth in the scientific literature in relation to the application of AI in education (AIEd). This study aims to present multiple perspectives on the development of AIEd in terms of relevant grants, conferences, journals, software tools, article trends, top issues, institutions, and researchers to provide an overview of AIEd for its further development and implementation. With this study, we contribute to the research field by enabling educators and scholars to understand the status and development of relevant grants and publications concerning AIEd. Also, findings concerning active actors can help educators and scholars identify the active researchers and institutions in the research on AIEd. Furthermore, researchers and educators are able to identify relevant journals and be more aware of major issues in AIEd studies. In addition, we also highlight the significance and necessity of the launch of the new Elsevier journal AIEd-related journal named *Computers & Education: Artificial Intelligence*.

1. Introduction

Technologies have a significant impact on today's society and human beings. They have greatly changed the way we think, feel, act, and how we communicate and interact with each other. Meanwhile, technological advancements are rapidly and tremendously changing the ways in which we learn and teach. Consequently, the landscape of education is gradually changing.

Especially in recent years, with the advance and prevalence of interactive technologies, their applications within and outside of classes are steadily gaining in popularity. Such technologies aim to adapt to learners' diverse needs and situations, as well as to enhance competitiveness in the global education market. Students utilize tablets rather than copybooks, and teachers use various educational tools and applications such as Google Classroom and Moodle. Also, a great number of Massive Open Online Course (MOOC) platforms like Class Central, EdX, and Coursera are increasingly available for online learning. The increasing number of active MOOC users indicates the wide acceptance and appreciation of digital and distance learning technologies. Additionally, there are MOOCs that even provide learners with opportunities to obtain official degrees from distinguished universities and institutes.

Intrinsically, the future of education is, to a large degree, related to the advances in novel intelligent technologies and the rapid development of computational capacities. Hence, the educational community is facing the opportunity and challenge brought by the continuously developing artificial intelligence (AI) technologies, which can potentially and fundamentally change the structure, operation, and governance of educational institutes (Popenici & Kerr, 2017).

The work in AI is dedicated to addressing cognitive problems that are commonly related to human intelligence. The educational sector is undeniably significantly influenced by AI. AI applications in education (AIEd) are widely used by learners and educators nowadays and involve various tools and applications, for example, intelligent tutoring systems, teaching robots, and adaptive learning systems. AIEd supports learning occurring in both traditional classes and workplaces by combining AI and various learning sciences, for example, education, psychology, linguistics, and neuroscience, aiming at stimulating and advancing the development of AI-driven educational applications featuring flexibility, personalization, and effectiveness(Luckin, Holmes, Griffiths, & Forcier, 2016).

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To present a multi-perspective overview of the development of AIEd, this study examines relevant grants, conferences, journals, software tools, article trends, top issues, institutions, and researchers. Such an investigation is important for the following reasons. First, scholars and their projects largely depend on funding as the main research driver, and every year considerable financial resources are invested in facilitating scientific activities. The trend analysis of funding in a particular field is essential in terms of examining whether the field is of great importance and research interest, and has great potential to make advances. Such analysis is also helpful for decision-makers in the determination of effective funding policies in (re)allocating resources to the most promising research fields and projects (Ebadi & Schiffauerova, 2016). Second, analysis of trends in scientific publications in a particular field can reflect the evolution of production and interest, as well as global scientific progress and development tendency in the field (Loonen et al., 2007; Ye et al., 2018). Generally, the growth in the number of publications usually reflects an expansion of the global science system concerning the field (Aksnes et al., 2016). Third, analysis of conferences and journals helps researchers and students identify relevant publication sources with an international scope (Song, Chen, Hao, Liu, & Lan, 2019). Fourth, the analysis of institutions and researchers helps scholars identify major contributors to the field with the most potential to develop and share research (Mair & Reischauer, 2017). Furthermore, analysis of research issues helps identify the foci of research-based publications in a particular field in the past, current, and future (Snyder, 2011). In addition, a survey of relevant software and products provides a basic understanding of the market size concerning a particular field, as well as what types of products have been commonly invested in and what can be further improved. Accordingly, there are six research questions we aim to answer:

RQ1: What was the trend of grants in relation to AIEd?
RQ2: What were the major conferences and journals related to AIEd?
RQ3: What were the major software tools concerning AIEd?
RQ4: What was the trend of AIEd publications?
RQ5: What were the top frequently used keywords in AIEd publications?
RQ6: What were the active institutions and researchers in the field of AIEd?

This study contributes to the research field in several aspects. First, it enables educators and scholars to understand the status and development of relevant grants and publications concerning AIEd. Second, it helps educators and scholars identify the active researchers and institutions in the research on AIEd. Third, researchers and educators are able to identify relevant journals dealing with AIEd to find channels to make contributions to the community. Fourth, it enables educators and scholars to be more aware of the top keywords being used in AIEd studies, which is especially helpful when they are making decisions about which types of issues to deal with. In addition, this research helps educators and scholars identify relevant software tools concerning the application of AI for educational purposes.

The rest of this article is outlined as follows. The data collection and analyses related to characteristics concerning AIEd are described in "Material and methods" section. The "Results" section presents the results and findings concerning grants, conferences, journals, software tools, article trends, top keywords, institutions, and researchers in relation to AIEd. In addition, we also introduce a newly developed AIEd-related Elsevier journal named *Computers & Education: Artificial Intelligence*. In the "Discussion" section, we provide a discussion on the findings and highlight the significance and necessity of the launch of the new journal. Finally, this study ends with a "Conclusion" section to summarize the major content and findings of our study.

2. Material and methods

To analyze the recent funding trends related to AIEd, we collected funding data concerning educational technology funded by several well-known international funding bodies, including the National Science Foundation,[1] the Australia Research Council,[2] the European Union fp7[3] and h2020 projects,[4] as well as the National Natural Science Foundation of China.[5] It should be noted that for the Australia Research Council, grants that are related to educational systems were considered. For each of the above funding bodies, the collected relevant grants were organized by funding year to calculate the annual number of grants, so as to depict how the number of grants evolved across time.

Table 1
Search query in Web of Science.

TS	(("artificial intelligence" OR "machine intelligence" OR "artificial neural network*" OR "machine learning" OR "deep learn*" OR "natural language process*" OR "robotic*" OR "thinking computer system*" OR "expert system*" OR "evolutionary computation" OR "hybrid intelligent system*") AND ("education*" OR "teaching"))
Language	English
Article types	"PROCEEDINGS PAPER" OR "ARTICLE"
Time span	1999–2019
Indexes	SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI

We collected data from the Web of Science (WoS) using the advanced search query provided in Table 1. We chose the WoS for the following reasons. First, WoS is the most popularly used database for bibliometric studies (Roemer & Borchardt, 2015). Second, the published studies in the WoS were generally subjected to the meticulous and peer-review process, ensuring high-quality publications. Additionally, in comparison to other databases, the bibliographic data collected from the WoS were more consistent and normative in expression. The search terms were determined by referring to previous studies (e.g., Heradio et al., 2016; Martí-Parreño et al., 2016; Tran et al., 2019, 2019a). The search was conducted on October 3, 2020. A total of 9560 records were collected. We analyzed the AIEd publication and citation trends and the major keywords being used, as well as the major publication sources, institutions, and countries/regions to understand the research status of AIEd research (e.g., Chen & Xie, 2020, Chen et al., 2018).

To identify the active researchers in the field of AIEd, we used citations provided by Google Scholar. Specifically, we included authors with a research interest in AIEd who had more than 1000 citations as of August 3, 2020.

To identify the top frequently adopted keywords in the 9560 AIEd publications, we collected the author-defined keywords as well as the keywords provided by the WoS (i.e., WoS Keywords Plus) for each publication and calculated their frequencies to determine those that were most commonly used.

We additionally provided an overview of some relevant journals, conferences, and software tools related to AIEd.

3. Results

3.1. Article and citation trends

The trend for publications from 1999 to 2019 is illustrated in Fig. 1. In general, the number of AIEd publications showed a growing trend over the studied years. The evolution of the published article output generally followed three stages. The first stage was from 1999 to 2002, with a comparatively slow growth tendency. The second stage featured a steady

[1] https://www.nsf.gov/.
[2] https://www.arc.gov.au/.
[3] https://ec.europa.eu/research/fp7/index_en.cfm.
[4] https://ec.europa.eu/programmes/horizon2020/en/h2020-sections-projects.
[5] http://www.nsfc.gov.cn/english/site_1/index.html.

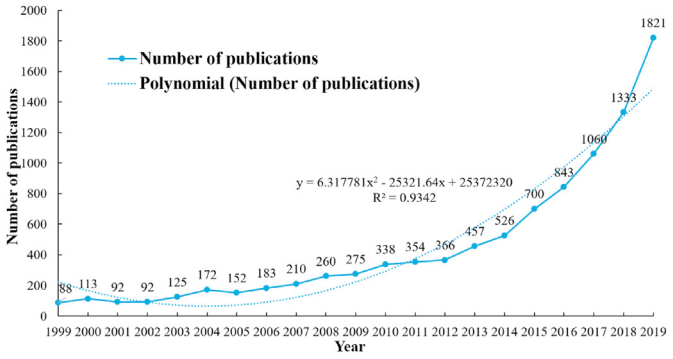


Fig. 1. Publication trend of AI in education.

growth tendency from 2003 to 2011. The third stage was a fast growth period from 2012 to 2019. The developmental trend of annual citations received by the studied publications (see Fig. 2) was quite similar, with a dramatic growing tendency during the past two decades. Figs. 1 and 2 show that 2019 was the most prolific year with the most publications and citations. Such results demonstrate that increased attention and interest in AI's application in educational contexts around the world led to a boost in publications about AIEd, with more than 74% of the literature being published between 2012 and 2019, leading to the conclusion that this is a rapidly emerging field of research.

3.2. Top subjects

The sample documents covered a total of 209 WoS subject categories. The research on AIEd covered a wide variety of themes and disciplines. The top 30 subject categories are displayed in Table 2, which shows that EDUCATION EDUCATIONAL RESEARCH ranked first with 2128 publications, constituting 22.26%, followed by COMPUTER SCIENCE ARTIFICIAL INTELLIGENCE with 1813 publications, and ENGINEERING ELECTRICAL ELECTRONIC with 1788 publications. COMPUTER SCIENCE THEORY METHODS, EDUCATION SCIENTIFIC DISCIPLINES, COMPUTER SCIENCE INTERDISCIPLINARY APPLICATIONS, and ROBOTICS were also relevant subject categories. AIEd is a multi-disciplinary area with major coverage of computer science and education (Kandlhofer et al., 2016). Additionally, since AI is itself multi-disciplinary, it has close interaction with various disciplines such as psychology and neuroscience.

3.3. Conferences and journals

1) Top publication sources

The studied 9560 AIEd publications were distributed across a total of 5240 sources, among which the top 30 are shown in Table 3. Among the listed sources, Lecture Notes in Computer Science published the most AIEd studies, with 222 publications (2.32%), followed by International

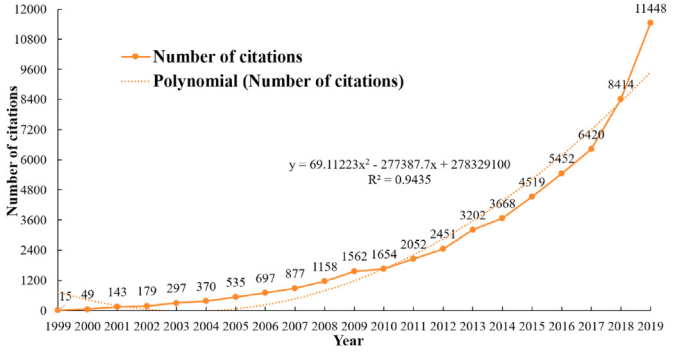


Fig. 2. Citation trend of AI in education.

Table 2
Top WoS categories.

WoS category	A	%
EDUCATION EDUCATIONAL RESEARCH	2128	22.26%
COMPUTER SCIENCE ARTIFICIAL INTELLIGENCE	1813	18.96%
ENGINEERING ELECTRICAL ELECTRONIC	1788	18.70%
COMPUTER SCIENCE THEORY METHODS	1309	13.69%
EDUCATION SCIENTIFIC DISCIPLINES	1290	13.49%
COMPUTER SCIENCE INTERDISCIPLINARY APPLICATIONS	1265	13.23%
ROBOTICS	1070	11.19%
COMPUTER SCIENCE INFORMATION SYSTEMS	919	9.61%
AUTOMATION CONTROL SYSTEMS	603	6.31%
ENGINEERING MULTIDISCIPLINARY	560	5.86%
COMPUTER SCIENCE SOFTWARE ENGINEERING	336	3.52%
SURGERY	312	3.26%
TELECOMMUNICATIONS	293	3.07%
COMPUTER SCIENCE CYBERNETICS	290	3.03%
ENGINEERING MECHANICAL	201	2.10%
HEALTH CARE SCIENCES SERVICES	153	1.60%
MEDICAL INFORMATICS	152	1.59%
SOCIAL SCIENCES INTERDISCIPLINARY	138	1.44%
COMPUTER SCIENCE HARDWARE ARCHITECTURE	137	1.43%
UROLOGY NEPHROLOGY	123	1.29%
ENGINEERING BIOMEDICAL	121	1.27%
ENGINEERING MANUFACTURING	117	1.22%
OPERATIONS RESEARCH MANAGEMENT SCIENCE	115	1.20%
ENGINEERING INDUSTRIAL	114	1.19%
INFORMATION SCIENCE LIBRARY SCIENCE	106	1.11%
EDUCATION EDUCATIONAL RESEARCH	2128	22.26%
COMPUTER SCIENCE ARTIFICIAL INTELLIGENCE	1813	18.96%
ENGINEERING ELECTRICAL ELECTRONIC	1788	18.70%
COMPUTER SCIENCE THEORY METHODS	1309	13.69%
EDUCATION SCIENTIFIC DISCIPLINES	1290	13.49%

Note: A: publication count; %: the proportion of publications in the total corpus.

Conference on Education and New Learning Technologies (196 publications), and International Technology, Education and Development Conference (178 publications). It should be noted that 18 out of the 30 listed sources, and all of the top nine, were proceedings. Furthermore, among the 2358 publications published by the 30 sources, about 75% were proceedings papers. This indicates that, currently, the major sources for researchers to make contributions to AIEd research are proceedings and conferences, which reflects the need to launch more international journals with a particular focus on AI in education, so as to increase the number of channels for scholars to make contributions to this field.

2) Examples of other active journals and conferences in AIEd

We further list here examples of other very relevant journals and conferences to AIEd research. There are four Elsevier journals which publish research papers on various facets of technology-enhanced education, including Computers & Education, The Internet and Higher Education, Computers and Composition, and System, as well as several non-Elsevier scholarly journals that have partially overlapping topic coverage with AIEd, as shown in Table 4.

Additionally, the following conferences, ICWL, ICALT, SIGITE, TEEM, AIED, ITS, L@S, EDM are quite relevant to AIEd. The statistics of submissions and accepted full papers are shown in Table 5.

3.4. Countries/regions, institutions, and researchers

The major contributors, including countries/regions and institutions for the publication of the 9560 AIEd studies, were identified. The top 20 countries/regions are presented in Table 6, and the top 30 institutions are listed in Table 7. The most prolific country in the AIEd research was the USA with 2581 publications, constituting 27%, followed by China (751 publications, 7.86%), and Spain (607 publications, 6.35%). In terms of prolific institutions, the top was the University of California System with 146 publications, constituting 1.53%, followed by the University System of Georgia (103 publications, 1.08%), and Pennsylvania Commonwealth

Table 3
Top publication sources.

Publication sources	Type	A	%
Lecture Notes in Computer Science	Proceedings	222	2.32%
International Conference on Education and New Learning Technologies	Proceedings	196	2.05%
International Technology, Education and Development Conference	Proceedings	178	1.86%
Advances in Intelligent Systems And Computing	Proceedings	166	1.74%
Lecture Notes in Artificial Intelligence	Proceedings	151	1.58%
Frontiers in Education Conference	Proceedings	147	1.54%
International Computing Education Research conference	Proceedings	110	1.15%
ASEE Annual Conference & Exposition	Proceedings	89	0.93%
Communications in Computer and Information Science	Proceedings	88	0.92%
International Journal of Engineering Education	Journal	84	0.88%
IEEE Transactions on Education	Journal	72	0.75%
Computers & Education	Journal	63	0.66%
Procedia Social and Behavioral Sciences	Proceedings	63	0.66%
IEEE Global Engineering Education Conference	Proceedings	62	0.65%
Computer Applications in Engineering Education	Journal	57	0.60%
Advances in Social Science Education and Humanities Research	Proceedings	50	0.52%
The International Conference on Robotics and Automation	Proceedings	50	0.52%
Society of Photo-optical Instrumentation Engineers	Proceedings	50	0.52%
Frontiers in Artificial Intelligence and Applications	Journal	49	0.51%
eLearning and Software for Education	Proceedings	46	0.48%
Surgical Endoscopy and Other Interventional Techniques	Journal	45	0.47%
Procedia Computer Science	Journal	44	0.46%
Expert Systems With Applications	Journal	42	0.44%
IFAC-PapersOnLine	Journal	42	0.44%
IEEE Access	Journal	41	0.43%
International Conference on Advanced Learning Technologies	Proceedings	31	0.32%
Computers in Human Behavior	Journal	30	0.31%
Proceedings of IEEE International Conference on Systems, Man and Cybernetics	Proceedings	30	0.31%
International Conference on Robot and Human Interactive Communication	Proceedings	30	0.31%
International Journal of Emerging Technologies in Learning	Journal	30	0.31%

Note: A: publication count; %: the proportion of publications in the total corpus.

System of Higher Education (101 publications, 1.06%). In addition, the active researchers in the field of AIEd are listed in Table 8.

3.5. Top keywords used in AIEd publications

The most frequently used keywords in the 9560 AIEd publications are listed in Table 9, including "education" (appearing in 1001 publications, constituting 10.47%), "machine learning" (80, 8.37%), "robotics" (737, 7.71%), "artificial intelligence" (667, 6.98%), "deep learning" (359, 3.76%), "system" (285, 2.98%), and "educational robotics" (277, 2.90%). It is worth noting that AI technologies such as machine learning, neural network, and natural language processing (NLP) were frequently mentioned in the AIEd studies. This indicates a wide concern with the application of AI technologies to resolve issues related to teaching and learning among scholars.

We further visualized the evolution of keywords used in AIEd studies, as shown in Fig. 3. From the results, it can be found that there were some keywords receiving increasing attention from AIEd scholars, for example, "classification," "stem," "computational thinking," "educational data mining," "neural network," "big data," "moocs," and "support vector machine (SVM)."

3.6. Impactful studies

We further identified impactful AIEd studies based on total citations and (TC) annual citations (C/Y) (Chen et al., 2018). As shown in

Table 4
Relevant journals in AIEd.

Names	Established year	IF (2019)	Publication frequency (issues/year)
Computers & Education	1976	5.296	10
The Internet and Higher Education	1998	6.566	4
Computers and Composition	1983	NA	4
System	1973	1.979	4
Educational Technology Research and Development	1953	2.303	6
International Journal of Artificial Intelligence in Education	2013	NA	4
International Journal of Computer-Supported Collaborative Learning	2006	4.028	4
International Journal of STEM Education	2014	1.85	1
Journal of Computers in Education	2014	NA	4
Journal of Computing in Higher Education	1989	2.271	2
Journal of Science Education and Technology	1992	1.644	6
Learning Environments Research	1998	NA	2
TechTrends	1985	NA	6
Technology, Knowledge and Learning	1996	NA	3
ACM Transactions on Computing Education	2001	1.721	4
IEEE Transactions on Education	1963	1.855	4
IEEE Transactions on Learning Technologies	2008	2.714	4
Journal of Computer Assisted Learning	1985	2.126	6
International Journal of Distance Education Technologies	2003	NA	4
British Journal of Educational Technology	1970	2.951	6
The International Review of Research in Open and Distributed Learning	2000	2.297	5
Journal of Educational Technology & Society	1998	NA	4
Education and Information Technologies	1996	NA	6
Educational Technology Research and Development	1953	2.303	6
Language Learning & Technology	1998	2.473	3
Interactive Learning Environments	1990	1.938	8
Turkish Online Journal of Educational Technology	2002	NA	4
Australasian Journal of Educational Technology	1985	1.956	6
Learning, Media and Technology	1975	2.547	4
Computer Assisted Language Learning	1990	2.642	8
Journal of Educational Data Mining	2009	NA	3

Table 10, the study by Dautenhahn (2007) appeared in both ranking lists. In this study, the author addressed dimensions of HRI. For one thing, the author investigated how cognitive robots could be used as a companion to develop robot behavior rules acceptable to the human being. For another thing, the author discussed the potential use of robots for educational or therapeutic purposes among children with autism. The study by Lété et al. (2004) introduced MANULEX, a web-accessible database which provided grade-level word frequency lists. MANULEX had the potential to be used to study language development through stimuli selection with the basis of precise frequency norms and to serve as an information source on NLP to facilitate children's writing. In the study by García et al. (2007), Bayesian networks was examined to identify

Table 5
The statistics of submissions and accepted full papers of relevant conferences in AIEd.

Conference names	Abbrev.	2016	2017	2018	2019
International Conference on Web-based Learning	ICWL	19/ 110	13/ 56	11/ 37	15/ 60
IEEE International Conference on Advanced Learning Technologies	ICALT	72/ 259	55/ 252	52/ 231	41/ 165
ACM Special Interest Group for Information Technology Education Conference	SIGITE	26/ 67	23/ 58	23/ 57	23/ 57
International Conference on Learning Analytics and Knowledge	LAK	36/ 116	36/ 114	35/ 115	42/ 115
International Conference on Technological Ecosystems for Enhancing Multiculturality	TEEM	165/ 235	83/ 109	151/ 243	N/A
International Conference on Artificial Intelligence in Education	AIED	NA	36/ 121	45/ 192	45/ 177
ACM Conference on Learning at Scale	L@S	18/ 79	14/ 105	N/A	
International Conference on Educational Data Mining	EDM	14/ 109	18/ 72	23/ 144	22/ 105

Table 6
Top countries/regions.

Country/Region	A	%
USA	2581	27.00%
China	751	7.86%
Spain	607	6.35%
England	576	6.03%
Germany	397	4.15%
Italy	340	3.56%
Japan	332	3.47%
Australia	327	3.42%
Canada	320	3.35%
India	319	3.34%
Brazil	250	2.62%
Romania	228	2.39%
Greece	213	2.23%
Russia	213	2.23%
Turkey	198	2.07%
France	177	1.85%
Taiwan	175	1.83%
Portugal	174	1.82%
South Korea	156	1.63%
Malaysia	145	1.52%

Note: A: publication count; %: the proportion of publications in the total corpus.

learners' learning styles in an e-learning system. Specifically, the researchers modeled various aspects of learner behavior, based on which learning styles were inferred. Through evaluation, it was found that the proposed Bayesian model was promising in the detection of learners' learning styles. Thomaz and Breazeal (2008) experimentally examined the assumption that human-given rewards were compatible with reinforcement learning reward signals by analyzing real-time human teaching behavior that happened while untrained subjects were teaching simulated reinforcement learning robots to conduct new tasks. Results demonstrated that it was essential to understand human-teacher/robot-learner partnerships so as to design algorithms supporting teaching and meanwhile improving robots' learning behavior. In the study by Lykourentzou et al. (2009), an innovative dropout prediction approach was proposed by using machine learning algorithms (i.e., feed-forward neural networks, SVMs, and probabilistic ensemble simplified fuzzy ARTMAP). Researchers conducted evaluation from the perspectives of accuracy, sensitivity, and precision, demonstrating the effectiveness of the proposed approach for predicting

Table 7
Top institutions.

Institutions	A	%
University of California System	146	1.53%
University System of Georgia	103	1.08%
Pennsylvania Commonwealth System of Higher Education	101	1.06%
Carnegie Mellon University	100	1.05%
State University System of Florida	95	0.99%
University of Texas System	81	0.85%
Harvard University	80	0.84%
University of London	77	0.81%
Massachusetts Institute of Technology	73	0.76%
Stanford University	69	0.72%
University of Southern California	64	0.67%
University of North Carolina	55	0.58%
Universidad Nacional de Educación a Distancia	54	0.57%
Georgia Institute of Technology	53	0.55%
Arizona State University	50	0.52%
Polytechnic University of Bucharest	50	0.52%
University of Patras	50	0.52%
University of Pittsburgh	50	0.52%
Johns Hopkins University	49	0.51%
Institute for Systems and Computer Engineering, Technology and Science	48	0.50%
Arizona State University Tempe	47	0.49%
University of Pennsylvania	46	0.48%
University of Toronto	46	0.48%
United States Department of Defense	45	0.47%
National Center for Scientific Research	44	0.46%
Ecole Polytechnique Federale De Lausanne	42	0.44%
University of Illinois System	42	0.44%
University of Michigan System	42	0.44%
Pennsylvania State University	41	0.43%
University of Michigan	41	0.43%

Note: A: publication count; %: the proportion of publications in the total corpus.

dropout. A comparative study was presented by Costa, Fonseca, Santana, de Araújo, and Rego (2017) to examine the potential of educational data mining techniques (i.e., SVMs, decision trees via J48, neural network, and Naive Bayes) for failing prediction in an introductory programming course. Results indicated that data mining techniques were effective for the task, and SVMs outperformed others with statistical significance. Whitehill et al. (2014) developed a real-time automatic approach for engagement detection based on learners' facial expressions with the use of GentleBoost with Box filter features, SVMs with Gabor features, and multinomial logistic regression with expression outputs from computer expression recognition toolbox (Littlewort et al., 2011). Results indicated that machine learning contributed to the effectiveness of real-time automated engagement detectors. With the basis of a temporal prediction mechanism, Xing and Du (2019) aimed at constructing a dropout prediction model by using deep learning techniques. In addition to achieving high accuracy in dropout prediction, their proposed model was capable of personalizing and prioritizing intervention for at-risk MOOCs learners based on individual dropout probabilities. Hussain et al. (2019) aimed at predicting what difficulties students would encounter in an e-learning system by analyzing logfiles with the use of machine learning (i.e., artificial neural networks, SVMs, logistic regression, Naive Bayes, and decision trees). Through evaluation, it was found that artificial neural networks and SVMs outperformed others in difficulty prediction.

3.7. Grants

The number of grants, as well as the funding amount for educational technology from the National Science Foundation, the Australia Research Council (with educational system-related funding considered), the European Union fp7 and h2020 projects, as well as the National Natural Science Foundation of China, are shown in Fig. 4. In general, the funding of the educational technology area has enjoyed a growing tendency.

Table 8
Active researchers in AIEd (listed in alphabetical order).

Authors	Institution	Country/ Region
BARNES, Tiffany	North Carolina State University	USA
BITTENCOURT, Ig Ibert	Federal University of Alagoas	Brazil
BOUCHET, François	Sorbonne Université	France
BRUSILOVSKY, Peter	University of Pittsburgh	USA
BULL, Susan	University of Birmingham	UK
CHOU Chih-Yueh	Yuan Ze University	Taiwan
DIMITROVA, Vania	University of Leeds	UK
FOLTZ, Peter	University of Cordoba	Spain
GOLDBERG, Benjamin S.	Simulation and Training Technology Center	USA
GOWDA, Sujith	Alpha Data Labs, LLC	USA
HARLEY, Jason Matthew	McGill University	Canada
HAYASHI, Yusuke	Hiroshima University	Japan
HWANG, Gwo-Jen	National Taiwan University of Science and Technology	Taiwan
ISOTANI, Seiji	University of Sao Paulo	Brazil
JEURING, Johan	Utrecht University	the Netherlands
KOEDINGER, Ken	Carnegie Mellon University	USA
KUMAR, Vivekanandan Suresh	Athabasca University	Canada
LEE, Chien-I	National University of Tainan	Taiwan
LESTER, James	North Carolina State University	USA
LIN, Yen-Ting	National Pingtung University	Taiwan
LITMAN, Diane	University of Pittsburgh	USA
LUCKIN, Rose	University College London	UK
MATSUDA, Noboru	North Carolina State University	USA
MCCALLA, Gordon	University of Saskatchewan	Canada
MCQUIGGAN, Scott	SAS Institute, Inc.	USA
MILLÁN, Eva	Universidad de Málaga	Spain
MITROVIC, Antonija	University of Canterbury	New Zealand
MIZOGUCHI, Riichiro	Japan Advanced Institute of Science and Technology	Japan
MOSTOW, Jack	Carnegie Mellon University	USA
NKAMBOU, Roger	Université du Québec à Montréal	Canada
PELÁNEK, Radek	Masaryk university	Czech
REBOLLEDO-MENDEZ, Genaro	Tec de Monterrey	Mexico
RODRIGO, Ma. Mercedes	Ateneo de Manila University	Philippines
ROMERO, Cristobal	University of Cordoba	Spain
SABOURIN, Jennifer	North Carolina State University	USA
SHAW, Erin	University of Southern California	USA
TIAN, Feng	Xi'an Jiaotong University	China
VASSILEVA, Julita	University of Saskatchewan	Canada
WANG, Ning	University of Southern California	USA
WOOLF, Beverly Park	University of Massachusetts	USA
XIE, Haoran	Lingnan University	Hong Kong
YACEF, Kalina	University of Sydney	Australia
YANG, Stephen	National Central University	Taiwan
ZAPATA-RIVERA, Diego	Educational Testing Service	USA

3.8. Software tools

Table 11 presents some of the software tools that use AI technologies for educational purposes. Among the 30 listed software tools, it should be noted that 70% are used for language learning purposes, and 20% are related to mathematics learning. This indicates the popularity of applying AI technologies to facilitate language and mathematics learning. A possible reason for this may be the fact that most of the software are designed for primary and secondary students, for whom the learning of language and mathematics is a major task during their study. However, such results also reflect the lack of software designed for learning other types of subjects such as chemistry, literacy, and programming.

3.9. Introduction of Computers & Education: Artificial intelligence

The analysis of the publication sources indicates that most of the current AIEd studies are published by proceedings, so it is essential that more international AIEd-focused journals be launched to provide more contributing channels for scholars to share and publish their latest

Table 9
Top keywords used in AIEd publications.

Keywords	A	%	Keywords	A	%
education	1001	10.47%	behavior	68	0.71%
machine learning	800	8.37%	strategy	67	0.70%
robotics	737	7.71%	mechatronics	67	0.70%
artificial intelligence	667	6.98%	program	67	0.70%
deep learning	359	3.76%	achievement	66	0.69%
system	285	2.98%	feedback	66	0.69%
educational robotics	277	2.90%	environment	66	0.69%
natural language processing	268	2.80%	lego mindstorm	66	0.69%
expert system	253	2.65%	algorithm	64	0.67%
design	241	2.52%	internet of things	62	0.65%
performance	234	2.45%	educational robot	62	0.65%
e-learning	233	2.44%	active learning	61	0.64%
student	229	2.40%	information	61	0.64%
technology	221	2.31%	educational technology	61	0.64%
model	200	2.09%	diagnosis	61	0.64%
simulation	190	1.99%	arduino	60	0.63%
robot	190	1.99%	quality	60	0.63%
higher education	180	1.88%	learning approach	59	0.62%
classification	174	1.82%	teacher	57	0.60%
science	158	1.65%	tool	57	0.60%
neural network	154	1.61%	problem based learning	57	0.60%
intelligent tutoring system	153	1.60%	collaborative learning	56	0.59%
surgery	152	1.59%	fuzzy logic	56	0.59%
skill	150	1.57%	recognition	55	0.58%
human-robot interaction (HRI)	146	1.53%	augmented reality	55	0.58%
data mining	140	1.46%	optimization	54	0.56%
robotic surgery	139	1.45%	mathematics	53	0.55%
stem	133	1.39%	computer science education	53	0.55%
educational data mining	133	1.39%	stem education	52	0.54%
engineering education	133	1.39%	mobile robotics	52	0.54%
learning	131	1.37%	intervention	51	0.53%
children	130	1.36%	pedagogy	51	0.53%
knowledge	129	1.35%	information and communication technology	50	0.52%
virtual reality	126	1.32%	health	50	0.52%
learning analytics	125	1.31%	creativity	50	0.52%
computational thinking	117	1.22%	computer vision	50	0.52%
outcome	116	1.21%	simulator	50	0.52%
experience	114	1.19%	genetic algorithm	50	0.52%
validation	109	1.14%	robotics education	50	0.52%
programming	105	1.10%	learning style	49	0.51%
impact	103	1.08%	care	49	0.51%
big data	97	1.01%	blended learning	48	0.50%
curriculum	94	0.98%	game	48	0.50%
motivation	87	0.91%	remote lab	48	0.50%
teaching	87	0.91%	convolution neural network	47	0.49%
support vector machine	82	0.86%	classroom	46	0.48%
perception	82	0.86%	self-efficacy	46	0.48%
training	82	0.86%	sensor	45	0.47%
mobile robot	82	0.86%	robotic	45	0.47%
project based learning	81	0.85%	sentiment analysis	44	0.46%
ontology	80	0.84%	online	44	0.46%
management	80	0.84%	automation	44	0.46%
moocs	77	0.81%	cancer	44	0.46%
artificial neural network	76	0.79%	evaluation	44	0.46%
framework	75	0.78%	future	43	0.45%
surgical education	75	0.78%	clustering	43	0.45%
learning curve	75	0.78%	decision tree	43	0.45%
laparoscopy	75	0.78%	social media	42	0.44%
prediction	74	0.77%	social robot	42	0.44%

(*continued on next page*)

Table 9 (*continued*)

Keywords	A	%	Keywords	A	%
engagement	73	0.76%	network	42	0.44%
assessment	73	0.76%	risk	41	0.43%
internet	73	0.76%	engineering	41	0.43%
school	72	0.75%	university	41	0.43%
medical education	69	0.72%	online learning	41	0.43%
innovation	68	0.71%	emotion	40	0.42%

Note: Some of the search terms used in our data collection also appeared in this table as frequently used keywords in the studied 9560 AIEd studies. A: publication count; %: the proportion of publications in the total corpus.

research work. As a demonstration of the need for more AIEd-focused journals, we present the publication trends of two relevant existing Elsevier journals related to educational technology, namely, Computers & Education and The Internet and Higher Education (see Fig. 5). From the annual trends of the publications in these two journals, we can see that they have reached saturation and tend to control the number of publications rather than expanding the scale. However, referring to the trend of publications related to AIEd (see Fig. 1), the continuing and fast-growing trend in this area is obvious. Such a phenomenon further demonstrates the great need to set up a community specifically concerning AIEd to provide opportunities for scholars to share their research on AIEd and to facilitate the further development of AIEd. With this in mind, there is a newly developed Elsevier journal called *Computers & Education: Artificial Intelligence* (*CAEAI*), which was launched at the moment when the whole field of AIEd was undergoing dynamic development. *CAEAI* aims to offer a platform for scholars, practitioners, and educators to publish their studies, share their ideas, and highlight novel systems and pedagogical innovation on issues concerning AI's use for educational purposes. As a sister journal to Computers & Education, *CAEAI* specifically focuses on AI's application in education, for example, educational data analysis based on AI algorithms, e-learning system design based on AI, and learner emotion and behavior detection based on AI.

4. Discussions and conclusion

In this article, a multi-perspective study on the development of AIEd is presented, particularly in terms of grants, conferences, journals, software tools, institutions, and researchers. It was found that research in the AIEd field entered a stage of rapid development around 2012, and the numbers are still growing. The dramatic growth in citations received by the AIEd studies demonstrates the increasingly wide impact and influence of research on AIEd in academia. Such findings are consistent with previous reviews (e.g., Chen, Zou, & Xie, 2020, Chen, Zou, Cheng, & Xie, (2020), Yang, Hwang, & Sung, 2020, Xie, Chu, Hwang, & Wang, 2019) concerning educational technologies, uncovering increasing interest in research issues related to technology-related education. Results of subject analysis reveal a diverse interest in AIEd among scholars from a variety of fields. However, most of the AIEd studies (e.g., Huang, Hwang, & Chang, 2020, Chang, Kao, Hwang, & Lin, 2020, Abbas, Hammad, Hwang, Khan, & Gilani, 2020) were distributed in the EDUCATION EDUCATIONAL RESEARCH and COMPUTER SCIENCE ARTIFICIAL INTELLIGENCE categories, indicating its close relation to education and computer science, particularly AI technologies. This is consistent with the definition and aim of AIEd as to apply AI technologies for achieving various educational purposes (e.g., Chen, Xie, Zou, & Hwang, 2020, Hwang, Xie, Wah, & Gašević, 2020).

Table 10
Examples of impactful AIEd studies.

Studies	Journal	TC
Dautenhahn (2007)	Philosophical Transactions of the Royal Society B	357
Lété et al. (2004)	Behavior Research Methods Instruments & Computers	237
García et al. (2007)	Computers & Education	210
Thomaz and Breazeal (2008)	Artificial Intelligence	135
Lykourentzou et al. (2009)	Computers & Education	126
Studies	Journal	C/Y
Dautenhahn (2007)	Philosophical Transactions of the Royal Society B	27.5
Costa et al. (2017)	Computers In Human Behavior	22.3
Whitehill et al. (2014)	IEEE Transactions on Affective Computing	20
Xing and Du (2019)	Journal of Educational Computing Research,	20
Hussain et al. (2019)	Artificial Intelligence Review	19

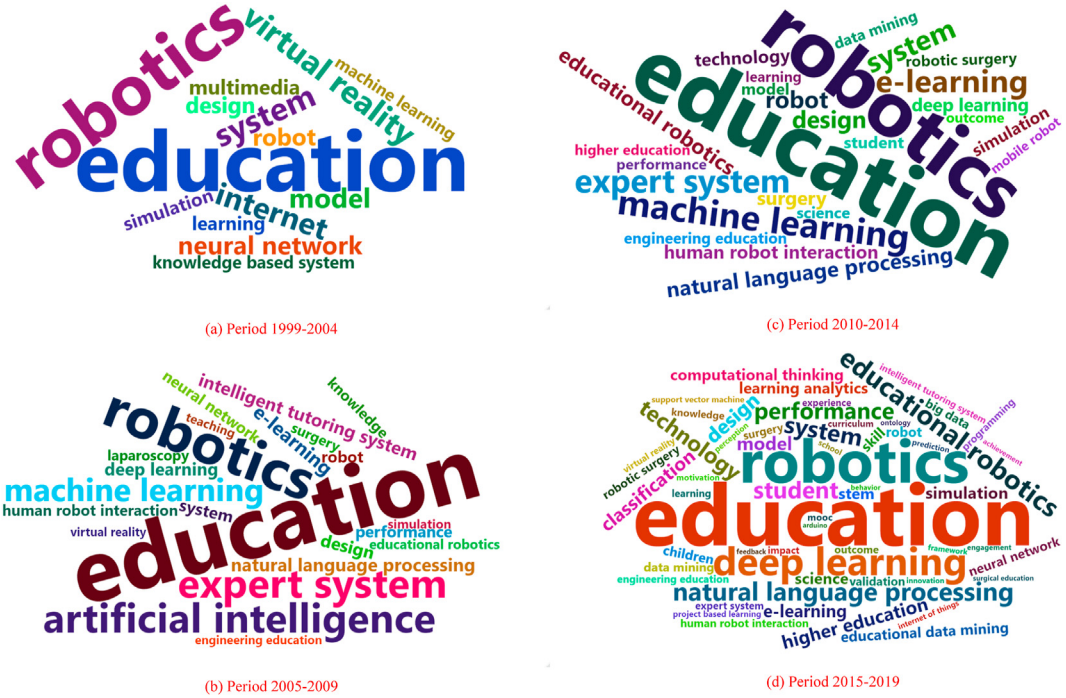


Fig. 3. Evolution of keywords used in AIEd studies.

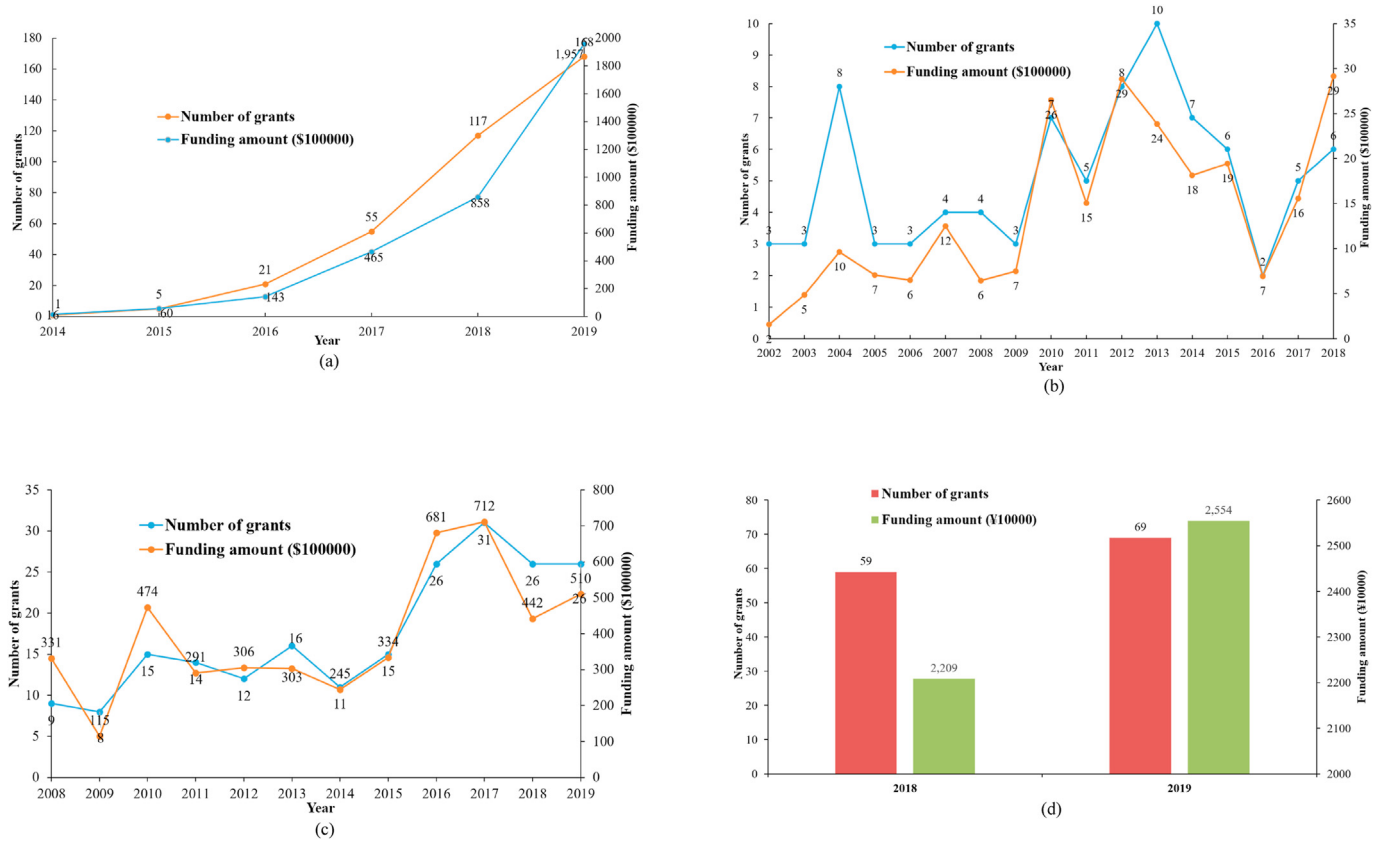


Fig. 4. Funding trends in educational technology of the National Science Foundation (a), the Australia Research Council (b), the European Union fp7 and h2020 projects (c), and the National Natural Science Foundation of China (d).

Results of the trend of grants in relation to educational technology from major bodies like the National Science Foundation and Australia Research Council indicate a significantly growing interest in and enthusiasm for this field. Governments in various countries/regions such as the United States and the United Kingdom have been furiously investing in the development of technological instruments that are incorporated with AI technologies in classroom settings, particularly in the recent few years. The education and scientific research issues in relation to AIEd are thus of increasing interest internationally, not only among educators and researchers but also among governments and educational institutions. For example, in 2018, the National Natural Science Foundation of China, for the first time, added the level-2 code F0701 of Education Information Science and Technology to encourage research on technology-enhanced education to facilitate the prospect of China's educational technology research.

Publication source analysis indicates that currently, conference proceedings dominate the publication of AIEd studies, which can be further confirmed by the relatively prevalent conferences in relation to the application of information technologies to resolve issues concerning teaching/learning, for example, the International Conference on Learning Analytics and Knowledge, the International Conference on Artificial Intelligence in Education, and the International Conference on Educational Data Mining. Comparatively, the contribution of journals in terms of publishing AIEd studies was not as significant as conferences. This may be due to the lack of journals that are particularly focusing on this area, as many of the AIEd journal articles were published in educational technology-related journals such as Computers & Education and Internet and Higher Education. However, the trend analysis of the publications in these two journals indicates that they tend to control the number of publications rather than expanding the scale. Thus, considering the continuing growth and development of research on AIEd, there is a great need for more academic journals with a particular focus on AI applications for educational purposes to be launched so as to provide opportunities for researchers to share their AIEd work and to facilitate the development of this field. In this sense, it is worth highlighting a recently launched journal, *CAEAI*, the objective of which is to afford a world-wide platform for researchers, developers, and educators to share their ideas and studies in relation to AIEd.

In terms of available educational software with the integration of AI technologies, much has been designed for mathematics and language learning. How AI technologies can facilitate the learning of other types of subjects such as chemistry and programming should be further investigated and turned into practice in the form of software tools. Actually, in academia, researchers have been working on these directions. For example, Serban et al. (2020) proposed a dialogue-driven intelligent tutoring system Korbit. Korbit, with the use of machine learning, NLP, and reinforcement learning, aimed at providing interactive and tailored learning experiences. To promote STEM learning, Korbit used a mixed-interface involving various elements such as multimedia, dialogue-driven exercises, concept mapping, and gamification. To relieve educators' workload in assisting learners, particularly novice learners, to fix programs, Miao et al. (2020) proposed PIPE, a deep learning method that was capable of predicting logical programming errors. The model seamlessly integrated a representation learning algorithm and a multi-label classification algorithm to obtain the latent feature of a program and to further predict the errors. Furthermore, there are scholars working on the development of novel applications for facilitating education based on commercial AI-related products and tools. For example, Ralston et al. (2019) proposed and developed a voice interactive and multilingual chatbot capable of responding to learners' mood, tone, and language with the use of IBM Watson Assistant, Tone Analyzer, and Language Translator. Experimental results of the analysis of responses concerning examination stress indicated the effectiveness of the chatbot in responding to learners. It is worth noting that the chatbot could be

Table 11
Examples of software tools used to apply AI technologies for educational purposes.

Name of app	Subject	Description	Website
Amira	literacy	Amira is a web-supported, AI-empowered educational application for K-3 literacy learning[1]	[1]https://www.amiralearning.com/
Carnegie Learning	math	Carnegie Learning is an AI-empowered app for helping learners cultivate deep conceptual understanding of math[2]	[2]https://www.carnegielearning.com/
Cognii	language	Cognii provides AI-driven educational tools for K-12 education, higher education, and corporate training[3]	[3]https://www.cognii.com/
CueThink	math	CueThink is an AI-driven app for math education that is able to assess learner knowledge based on an engaging peer-to-peer strategy[4]	[4]https://www.cuethink.com/
DreamBox Learning	math	DreamBox Learning is an AI-driven application for math education targeting elementary and middle school students[5]	[5]https://www.dreambox.com/
Edwin	language	Edwin provides AI-empowered services to learn foreign languages[6]	[6]https://edwin.ai/
Elemental Path	language	Elemental Path is an AI-driven product for learners to engage with real-time speech-driven communication[7]	[7]https://techcrunch.com/
GlobalEnglish	language	GlobalEnglish is an English learning application based on machine learning technologies[8]	[8]https://globalenglish.com/
Grammarly	language	Grammarly is a machine learning-driven app combined with various NLP technologies for improving writing[9]	[9]https://app.grammarly.com/
iTutorGroup	language	ITutorGroup is an AI-driven application for individualized and personalized language learning[10]	[10]http://www.itutorgroup.com/
KidSense	language	KidSense provides AI-driven applications for children's' language learning[11]	[11]https://kidsense.ai/
Lingco Classroom	language	Lingco Classroom is an AI-driven tool for personalized language learning [12]	[12]https://www.lingco.io/
Linguix	language	Linguix is an AI-driven tool for detecting and resolving spelling and grammar issues in writing[13]	[13]https://linguix.com/
MyGrammarCheck	language	MyGrammarCheck is a grammar checker based on machine learning technologies[14]	[14]https://www.mygrammarcheck.com/
Newsela	language	Newsela is an AI-driven tool for improving reading comprehension with rich article selection on various topics[15]	[15]https://newsela.com/
Noplag App	language	Noplag App is a writing assistance tool for plagiarism checking based on AI technologies [16]	[16]https://noplag.com/
Nuance	language	Nuance provides AI-driven speech recognition tools for students and faculty[17]	[17]https://www.nuance.com/
Pimsleur	language	Pimsleur is an AI-driven language learning app[18]	[18]https://www.pimsleur.com/
Querium	STEM	Querium is an AI-driven application for delivering customizable STEM tutoring courses to high school and college students[19]	[19]http://querium.com/
QuillBot	language	QuillBot is an AI-driven paraphrasing tool[20]	[20]https://quillbot.com/
Quizlet	language, science, math, history, and coding	Quizlet provides services ranging from the learning of foreign language, history, and science to standardized test preparation[21]	[21]https://quizlet.com/
Ready4	language	Ready4 provides services concerning test preparation and college admissions services startup based on AI technologies[22]	[22]https://www.ready4.com/
Rosetta Stone Language Learning	language	Rosetta Stone Language Learning is an augmented reality AI-empowered app for language learning[23]	[23]https://www.rosettastone.com
Symbolab	math	Symbolab is an AI-driven tool for mathematics education[24]	[24]https://www.symbolab.com/
Thinkster Math	math	Thinkster Math provides custom programs for individual learners based on AI technologies[25]	[25]https://hellothinkster.com/
TrueShelf	math	TrueShelf is AI-empowered adaptive math learning software for middle school and high school students[26]	[26]https://trueshelf.com/
VIPKID	language	VIPKID is an AI-empowered personalized language learning tool[27]	[27]https://www.vipkid.com
Yabla	language	Yabla provides personalized course materials based on AI technologies[28]	[28]https://www.yabla.com/
Maths AI App		Maths AI App is an AI-empowered online K-12 mathematics educational tool that is able to provide tailored learning materials[29]	[29]https://www.mathsai.com/
Talk2Learn	Math	Talk2Learn is an open-source AI-driven mathematics tutoring tool featuring revolutionary innovations such as an adaptive sequencer and voice recognition[30]	[30]https://www.italk2learn.com/

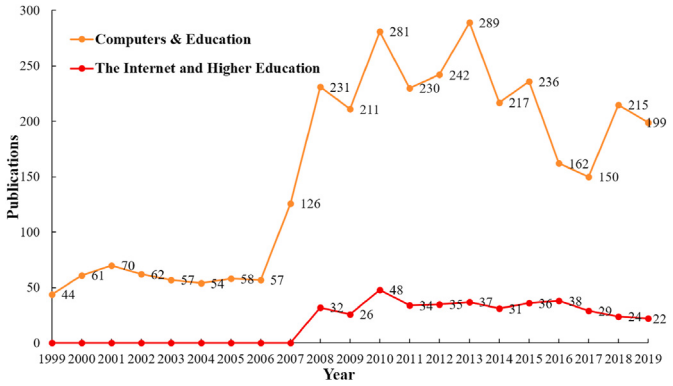


Fig. 5. Publication trends of Computers & Education and The Internet and Higher Education.

widely applied in different scenarios, for example, student information centers and mental health services. In addition, how to facilitate learning by using AI technologies has also received great interest from the industry. For example, IBM created IBM AI Education,[6] an immersive online professional learning suite, focusing on applying AI to promote learning outcomes.

Results of top keywords used in AIEd publications demonstrate a broad interest in technical issues such as the use of various AI technologies (e.g., machine learning, artificial neural network, convolution neural network, genetic algorithm, natural language learning, sentiment analysis, classification, and clustering) and different types of applications and tools developed mainly based on AI technologies such as mobile games, intelligent tutoring systems, educational robots, augmented reality, virtual reality, and serious games. Second, several educational strategies, such as project-based learning, collaborative learning, blended learning, problem-based learning, and mobile learning, are of great interest among AIEd researchers. Third, issues concerning learners' motivation, learning style, academic performance, computational thinking, perception, achievement, behavior, creativity, problem-solving, self-efficacy, emotion, critical thinking, and attitude are of great interest to AIEd scholars. Fourth, engineering education, higher education, mathematics education, language education, surgical education, robotics education, computer science education, STEM education, medical education, and science education are popular educational contexts to incorporate AI technologies to facilitate students' learning. Furthermore, there are several sub-fields in educational technology, such as learning analytics and educational data mining, which are frequently mentioned in AIEd studies. Although these sub-fields are closely related to AIEd, it is suggested that research pioneers should clarify the differences between each of the sub-fields with their unique research foci or issues to be highlighted. In addition, AIEd scholars have been interested in adopting AI technologies to facilitate the learning of students with special needs, such as those with dementia and autism spectrum disorders. However, although AIEd scholars have shown diverse interest in various perspectives and issues, little attention has been paid to educational theories, with only constructivism mentioned in the studies. It is advised that AIEd researchers should pay more attention to combine educational theory with AI technologies. In this way, their research can be more profound and theoretically sound with the guidance of authoritative and well-recognized theories.

To sum up, based on the results and findings, the increasing popularity and importance of AI applications for educational purposes has been confirmed and highlighted. AIEd scholars have shown great interest in various issues and perspectives in this field. However, educational theories seem to be absent. Also, in terms of educational software developed based on AI technologies, more attention should be paid to facilitate the learning of various subjects. Additionally, with the continually growing number of studies being conducted, a specialized journal, *CAEAI,* focusing specifically on the field, is needed and looks promising in terms of providing a platform for educators and researchers to share their AIEd studies and ideas. With this study, we hope to provide a brief overview of AIEd studies to facilitate readers' understanding of its active status and promising future.

Declaration of competing interest

As the research does not involve human participants directly, there is no need to seek ethical approval from a research review committee in the authors' affiliations. The data can be obtained by sending emails to the corresponding author. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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